AC Light Dimmer Module, 1 Channel, 3.3V/5V logic, AC 50/60hz, 220V/110V

DISCONTINUED | ROBOTDYN | SKU: RD-AC-DIM-1CH



Overview

The RobotDyn AC Light Dimmer Module is a single-channel TRIAC-based phase-cut controller designed to regulate AC power to resistive and some inductive loads. It allows microcontrollers like Arduino, ESP8266, ESP32, or STM32 to smoothly adjust brightness of incandescent or dimmable LED lamps, or control the speed of certain AC motors.

The high-voltage section is **galvanically isolated** from the logic side via an optocoupler, preventing any possibility of mains voltage reaching your microcontroller.

How It Works

The module uses **zero-cross detection** to synchronize switching with the AC waveform. When the AC voltage passes through zero, the ZC pin sends an interrupt to the microcontroller. The microcontroller then waits a calculated delay before firing the TRIAC via the PSM pin — the longer the delay, the less power reaches the load.

- At **50 Hz**, zero-crossing occurs 100 times per second
- At **60 Hz**, zero-crossing occurs 120 times per second

Pinout

PIN	FUNCTION	CONNECT TO
VCC	Logic power supply	3.3V or 5V
GND	Ground	GND
ZC	Zero-cross detection output	Digital input with interrupt support
PSM	TRIAC gate trigger	Any digital output

Zero-Cross Pin by Board

BOARD	ZC PIN	AVAILABLE OUTPUT PINS
Arduino UNO / Nano	D2	D0–D1, D3–D13
Arduino Mega	D2	D0–D1, D3–D70
Arduino Leonardo	D7	D0–D6, D8–D13
ESP8266	D1, D5, D7	D2, D6, D8
ESP32	Any GPIO with interrupt	Any GPIO

Compatible Loads

Works with:

- Incandescent lamps
- Dimmable LED lamps (marked as “dimmable”)
- Halogen lamps
- Heating elements
- Some AC motors (fans, drills)

Does NOT work with:

- Standard (non-dimmable) LED lamps
- Fluorescent / CFL lamps
- Devices with built-in electronic regulators
- Inductive loads without snubber circuit

Arduino Code Example

```
#include <RBDdimmer.h>

#define DIMMER_PIN 3
#define ZC_PIN 2

dimmerLamp dimmer(DIMMER_PIN);

void setup() {
  dimmer.begin(NORMAL_MODE, ON);
}

void loop() {
  for (int i = 0; i ⩽ 100; i++) {
    dimmer.setPower(i);
    delay(50);
  }
}
```

Install the library: RBDDimmer on GitHub

RBDdimmer Library Reference

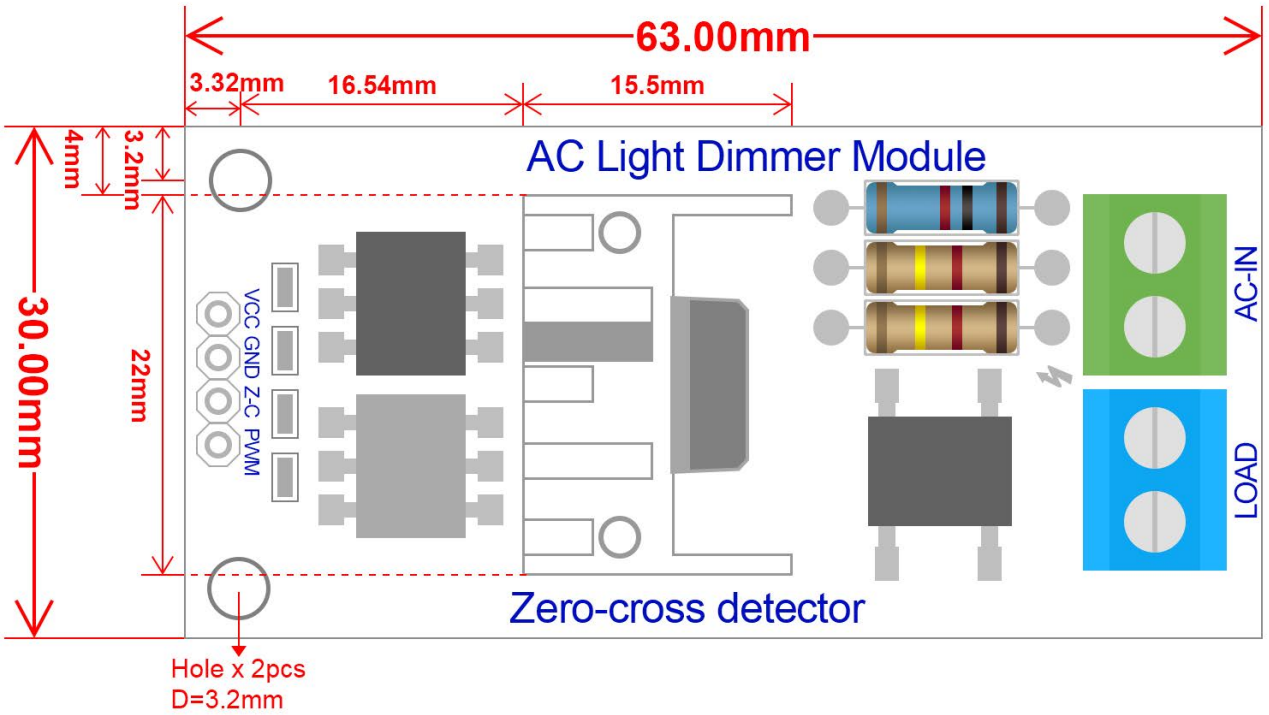
The library exposes 10 functions:

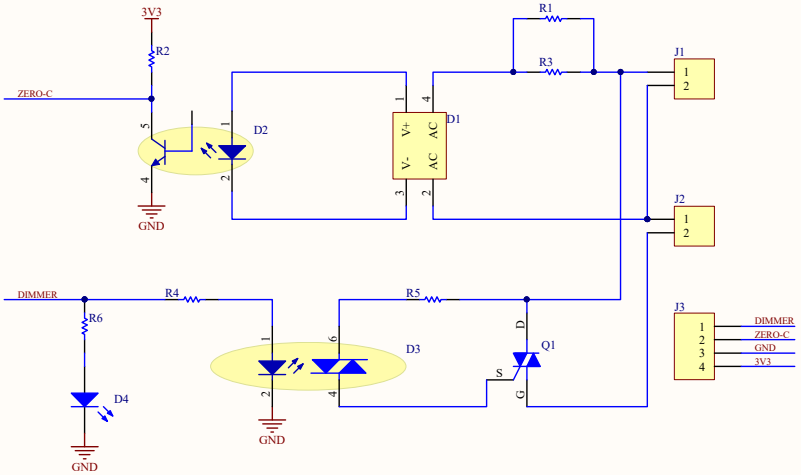
FUNCTION	PURPOSE
`dimmerLamp(psm_pin)`	Constructor — assigns the PSM (TRIAC trigger) pin. Example: `dimmerLamp dimmer(4);`
`begin(mode, state)`	Initializes ports, timer, and external interrupt. `mode` is `NORMAL_MODE` or `TOGGLE_MODE`; `state` is `ON` or `OFF`.
`setPower(value)`	Sets dimming level 0–100 (%). Example: `dimmer.setPower(90);`
`getPower()`	Returns current dimming value (0–100).
`setMode(mode)`	Switches between `NORMAL_MODE` (fixed level) and `TOGGLE_MODE` (auto-cycling).
`getMode()`	Returns current mode (`0` = NORMAL, `1` = TOGGLE).
`setState(state)`	Turns dimmer ON or OFF. Example: `dimmer.setState(ON);`
`getState()`	Returns current state (`0` = OFF, `1` = ON).
`changeState()`	Toggles state to its opposite (HIGH ⇔ LOW).

RobotDyn®
www.robotdyn.com

DIMENSION DIAGRAM

AC Light Dimmer Module, 1 Channel, 3.3V/5V logic, AC 50/60hz, 220V/110V





RobotDyn®
www.robotdyn.com

Mod-Dimmer-5A-1L